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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2024

ME1102-1 – MECHANICS OF MATERIALS

(For ME)

Duration: 3 Hours

Max. Marks: 100

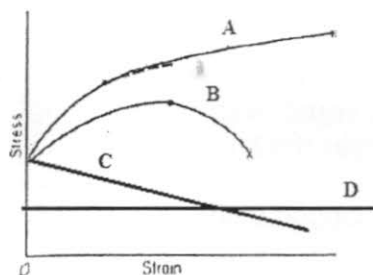
Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

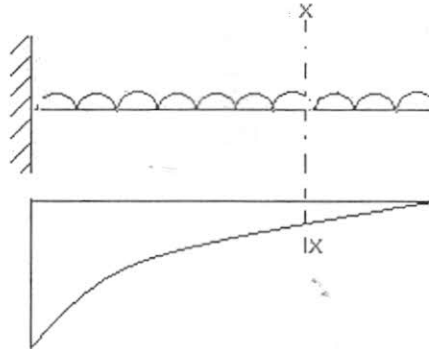
PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. The value of 50 MPa in terms of N/m^2 is
A) 5×10^7 B) 5×10^6
C) 50 D) 5×10^{10}
2. In the stress-strain diagram shown, which curve represents true stress strain curve?

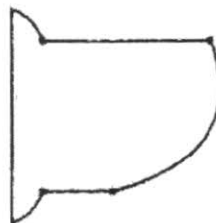


- A) curve A B) curve B
C) curve C D) curve D
3. A Mild Steel wire is 5 mm in diameter and 1 m long. If the wire is subjected to an axial tensile load of 10 kN find the extension of the rod. ($E = 200 \text{ GPa}$)
A) 25.5 mm B) 2.55 mm
C) .255 mm D) 255 mm
4. If d_1 and d_2 are the diameters of smaller and larger ends of a tapered rod of length 'L' subjected to a load 'P' at the end, the elongation will be
A) $4PL / \pi E d_1 d_2$ B) $4L / \pi E d_1 d_2$
C) $\pi E d_1 d_2 / 4PL$ D) $\pi E P d_1 d_2 / 4L$
5. E, G, K and μ are elastic modulus, shear modulus, bulk modulus and Poisson's ratio respectively. To express the stress strain relations completely for this material, at least
A) E, G and μ must be known B) E, K and μ must be known
C) Any two of the four must be known D) E, G, K and μ must be known
6. The Young's modulus of elasticity of a material is 2.5 times its modulus of rigidity. The Poisson's ratio for the material will be
A) 0.25 B) 0.33
C) 0.5 D) 0.75
7. In the analysis of thin cylinders, which of the stress being small is neglected?
A) hoop B) longitudinal
C) radial D) circumferential

8. At any point in a cylinder, Hoop stress and longitudinal stress are the two _____ stresses acting on it.
 A) tensile B) principal
 C) compressive D) shear
9. A shearing force having a downward direction to the right hand side of a section or upwards to the left hand of the section will be taken as
 A) negative B) unity
 C) zero D) positive
10. A bending moment causing _____ upwards will be taken as positive bending moment.
 A) concavity B) convexity
 C) curvature D) linearity
11. The graph shown below the beam in the figure represents it's _____



- A) Shear stress diagram B) Shear force diagram
 C) Bending moment diagram D) Bending stress diagram
12. A simply supported beam has equal over-hanging lengths and carries equal concentrated loads P at ends. Bending moment over the length between the supports
 A) is zero B) is a non-zero constant
 C) Varies uniformly from one support to the other D) Is maximum at mid-span
13. The bending equation is also called as _____ equation.
 A) Euler-Bernoulli's B) Pascal's
 C) Buckingham's D) Stoke's
14. The strength of the beam sections depends mainly on the
 A) bending moment B) section modulus
 C) bending stress D) polar modulus
15. A Cantilever beam of rectangular cross-section is 1m deep and 0.6 m thick. If the beam were to be 0.6 m deep and 1m thick, then the beam would
 A) Be weakened 0.5 times B) Be weakened 0.6 times
 C) Be strengthened 0.6 times D) Have the same strength as the original beam because the cross-sectional area remains the same
16. The shear stress distribution over a beam cross section is shown in the figure. The beam is of



- A) Equal flange I-Section B) unequal flange I-Section
 C) circular section D) T- section

17. A simply supported laterally loaded beam was found to deflect more than a specified value. Which of the following measures will reduce the deflection?
- A) Increase the area moment of inertia B) Increase the span of the beam
C) Select a different material having lesser modulus of elasticity D) Magnitude of the load to be increased
18. A simply supported beam with width 'b' and depth 'd' carries a central load W and undergoes deflection δ at the centre. If the width and depth are interchanged, the deflection at the centre of the beam would attain the value
- A) $d \delta / b$ B) $(d/b)^2 \delta$
C) $(d/b)^3 \delta$ D) $(d/b)^5 \delta$
19. The outside diameter of a hollow shaft is twice its inside diameter. The ratio of its torque carrying capacity to that of a solid shaft of the same material and the same outside diameter is
- A) 15/16 B) 3/4
C) 1/2 D) 1/16
20. The diameter of a shaft is increased from 30 mm to 60 mm, all other conditions remaining unchanged. How many times is its torque carrying capacity increased?
- A) 2 times B) 4 times
C) 8 times D) 16 times

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Draw the stress strain curve for mild steel indicating all salient points on it.	04	L*1	01	01
	b) Distinguish between				
	i. Ductility and Malleability	04	L2	01	01
	ii. Lateral Strain and longitudinal strain				
	c) Define Bulk Modulus. Derive the expression for relationship between elastic modulus and bulk modulus in terms of Poisson's ratio.	08	L2	02	01
2.	a) A tapering bar has diameter 'D ₁ ' at one end and it tapers uniformly to a diameter 'D ₂ ' at the other end for a length of 'L'. If modulus of elasticity of the material is 'E', Determine its change in length when subjected to an axial force 'P'	08	L2	01	01
	b) A 500 mm long bar has rectangular cross-section 20mm × 40mm. This bar is subjected to				
	(i) 40 kN tensile force on 20 mm × 40 mm faces,				
	(ii) 200 kN compressive force on 20 mm × 500 mm faces, and				
	(iii) 300 kN tensile force on 40 mm × 500 mm faces.				
	Find the change in volume if $E = 2 \times 10^5 \text{ N/mm}^2$ and $\mu = 0.3$.	08	L3	02	01
3.	a) For the laboratory tested specimen the following data were obtained. Diameter of specimen=12.5mm, Length of specimen = 150mm, Extension under the load of 16.4kN = 0.1mm, Load at yield point = 43kN, Maximum load = 60kN, Final length of the specimen = 190mm, Neck diameter at failure = 8mm. Determine (i) Young's Modulus (ii) Yield stress (iii) Ultimate stress (iv) Percentage Elongation (v) Percentage reduction in area.	08	L3	01	01
	b) A bar of 30mm diameter is tested in tension. It is observed that when a load of 60kN is applied, the extension measured over a gauge length of 200mm is 0.1mm and contraction in diameter is 0.004mm. Find the Poisson's ratio and elastic constants E,G,K.	08	L3	02	01

Unit – II

4. a) Elaborate types of loads used in beams.
 b) Define Bending moment and Shear force diagrams. Discuss the sign conventions used for shear force and bending moments.
 c) The T-section shown in fig.4(c).is used as a simply supported beam over a span of 4 meters. It carries a uniformly distributed load of 8 kN/m over its entire span. Calculate the maximum tensile and compressive stresses occurring in the section.

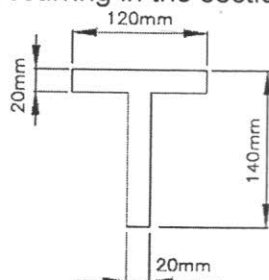


Fig.4(c)

02 L1 03 01
 06 L2 03 01

08 L3 04 02

5. a) A simply supported beam AB of 6m span is loaded as shown in fig. 5(a). Draw the shear force diagram with forces and moments.

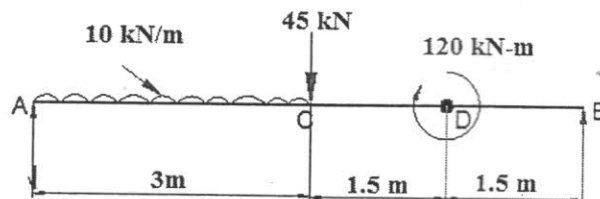


Fig. 5(a)

08 L3 03 02

- b) Show that for a rectangular cross section, shear stress distribution varies parabolically across depth.

08 L2 04 01

6. a) A simply supported beam AB of 6m span is loaded as shown in fig. 6(a). Draw the bending moment diagram if reaction force at A and F are 40kN each upward. Also indicate the point of contra flexure if any.

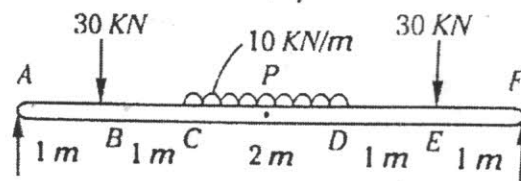


Fig. 6(a)

08 L3 03 02

- b) A beam of an 'I' section consists of 180mm x 15mm flanges and a web of 280mm depth x 15 mm thickness as shown in fig. 6(b). It carries a shearing force of 60kN at a section. Sketch the shear stress distribution across the depth of the section.

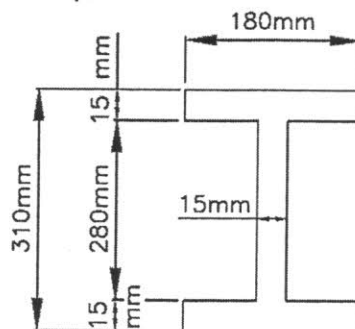


Fig. 6(b)

08 L3 04 02

Unit – III

7. a) Derive the expression for maximum deflection of a cantilever beam of length 'l' carrying a point load W at the free end.
- b) Find the maximum deflection and the maximum slope for the beam loaded as shown in fig. 7(b). Take $E=200 \text{ kN/mm}^2$, $I= 40 \times 10^6 \text{ mm}^4$.

08 L2 05 01

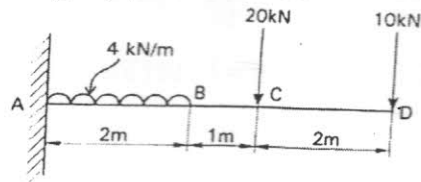


Fig. 7(b)

08 L3 05 02

8. a) Derive the relationship between angle of twist and shear stress in a solid circular shaft. Also mention assumptions made.
- b) A solid shaft rotating at 1000 rpm transmits 50 kW. Maximum torque is 20% more than the mean torque. Material of the shaft has the allowable shear stress of 50 MPa and Modulus of rigidity 80 GPa. Angle of the twist in the shaft should not exceed 1° in one-meter length. Determine the diameter of the shaft.

08 L2 05 01

08 L3 05 02

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
